

Nusret Efe Ucer

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EDUCATION

THE UNIVERSITY OF CHICAGO

Computer Science (B.S.) — Specialization in Computer Systems

Economics (B.A.) — Specialization in Data Science

GPA: 3.81/ 4.00

Awards: UChicago 2025 Summer Tech Showcase Runner-up; UChicago 2025 Winter Tech Showcase; UChicago Grant for Hackathon; Jeff Metcalf Award Recipient x2; IMC Prosperity 4 Finalist

Relevant Coursework: Honors Discrete Mathematics, TTIC31150 Graduate Abstract Linear Algebra and Probability, Introduction to Database Systems, Theory of Algorithms, Machine Learning, Statistical Methods and Applications, Investments, Introduction to Computer Security, Systems Programming I & II, Data Science for Computer Scientists, Foundations of Computer Networks

Programs: Trading Immersion Week Program, Financial Markets Program

Chicago, IL

Expected: June 2028

TECHNICAL SKILLS

Languages: Python, C, C++, Rust, Java, R, JavaScript, Tcl/Tk

Systems & Cloud: Linux, AWS (Lambda, SQS, DynamoDB, EventBridge, IAM, EC2), Azure, GCP, Git, TCP/IP, UDP, TLS, WireGuard

Frameworks & Libraries: PyQt, React, Flask, Node.js, NestJS, Streamlit

Data & ML: SQL, pandas, NumPy, Azure SQL, Databricks, MongoDB, Elasticsearch, Firebase, OpenAI API, Vertex AI

EXPERIENCE

Osmanli Investment Securities

Capital Markets & Trading Intern

Istanbul, Turkey

August 2026 – September 2026

- **Rotated across trading, portfolio management, and investment advisory desks** at a leading Istanbul-based brokerage (BIST: OSMEN), including live test-trading sessions with the Portfolio Management team.

Velocity Catalyst Fund

Pipeline/Data Engineering Intern

Chicago, IL

June 2026 – August 2026

- **Built the firm's technology stack from scratch on AWS as the sole engineer**, developing event-driven data pipelines using Lambda, SQS, SES, DynamoDB, Bedrock, and EventBridge.
- **Developed automations that accelerate deal flow and diligence**, and built an internal database integrated with the firm's CRM on AWS DynamoDB, tracking external and internal ventures and updates.

G LNK YC W25

ML Engineer (Part-Time) / ML Engineering Intern

Chicago, IL

June 2025 – June 2026

- **Built the company's database from zero**, scraping and unifying 300+ government healthcare datasets across CMS, NHS, and the French Ministry of Health; transformed 10^9 + Azure SQL records into 10^7 -scale Elasticsearch, Databricks, and MongoDB systems with custom schemas, indexing, ownership hierarchies, and cross-dataset links using OpenAI and Vertex AI.
- **Engineered the client-facing ETL pipeline** for non-healthcare expansion and data enrichment; optimized credit usage and parallelized execution across Azure VMs, multithreading, and asynchronous workflows, enabling same-day integrations and scaling coverage to approximately 92% of the global market across 170+ countries.
- **Built a full-stack search and analytics system** with Elasticsearch- and MongoDB-backed search, multi-criteria and regex filtering, a DataGrid frontend, and analysis pages with computed metrics and cost breakdowns.

UChicago Polarization-modulated Laser Satellite Experiment (PULSE-A)

Ground Station Software Engineer

Chicago, IL

October 2024 – Present

- **Built the PyQt Ground Station control system from scratch**, integrating GUI, backend logic, multithreaded and asynchronous process management, and real-time command and data handling for the Optical Ground Station telescope and laser-communication assembly.
- **Engineered TCP/UDP communication between Optical and RF Ground Stations over WireGuard/TLS**, achieving approximately 10–40 ms inter-city communication latency and sub-20 ms one-way confirmation for latency-sensitive tracking and control.

TECHNICAL PROJECTS

Maroon Exchange

June 2025 – August 2025

- **Co-built a peer-to-peer borrowing and lending platform** for University of Chicago students using React, Firebase authentication, and Azure SQL, with notification-driven matching, lender portfolios, and post-transaction ratings; runner-up at the UChicago Summer Tech Showcase, awarded \$5K.

“HandsIn” — HackTech 2025

April 2025

- **Built a sign-language conversational system** at Caltech HackTech after winning school grant funding; created a custom ASL dataset, trained a neural network from scratch, connected signing input to Gemini, and reconstructed responses in ASL through DeepMotion for a real-time signing avatar.

R19Q Mutation Analysis

March 2025 – May 2025

- **Analyzed the R19Q mutation in the SBDS protein** using a 20 ns molecular dynamics simulation on Midway3; performed RMSD/RMSF, Kabsch alignment, SASA, radius-of-gyration, hydrogen-bond, and time-series analyses to demonstrate mutation-driven domain dissociation associated with Shwachman–Diamond syndrome.

LEADERSHIP AND ACTIVITIES

UChicago Association for Computing Machinery (ACM)

President

Chicago, IL

October 2024 – Present

- **Led the AI/ML committee for ACM**, organizing events, workshops, and journal discussions on AI/ML literature, LLM fundamentals, model-building, prompt engineering, jailbreaking, and hyperparameter tuning.
- **Served as a core ACM board member**, overseeing event organization, interviewing new members, and participating in board meetings supporting chapter operations. Hosted Ayan Gupta (Microsoft), Maxim Allouch (Vera Health YCS24), Ce Zhang (together.ai Founder & CTO).